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Static Charge Detective

1. Will the objects below attract or repel each other?

1. _____ **2.** _____ **3.** _____

4. _____ **5.** _____ **6.** _____

2. Use the electrostatic series (see reverse) to answer the following questions:

   a) As you take clothes out of the dryer, a wool sock clings to a silk shirt.

      The wool sock has become _____ charged.

      The silk shirt has become _____ charged.

   b) You use a paper towel to rub off some dirt on a glass window.

      The glass window becomes _____ charged.

      The paper towel becomes _____ charged.

c) You rub a balloon along a cat's back, causing the cat's fur to stand up.

The cat's fur has become _____ charged.

The balloon has become _____ charged.

d) You rub an ebonite rod with nylon.

The nylon becomes _____ charged.

The ebonite rod becomes _____ charged.

3. Imagine the following scenario:
   - Fabric A is used to rub a wooden rod.
   - A second piece of Fabric A is used to rub an ebonite rod.

It is observed that the wooden rod and the ebonite rod now attract one another.

What can you say about the <u>position</u> of Fabric A in the electrostatic series compared to the wood and ebonite? Explain your reasoning in the space below.

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Table 1 Electrostatic Series

Material	Charge tendency
human skin	+ (weaker tendency to gain electrons)
rabbit fur	
acetate	
glass	
human hair	
nylon	
wool	
cat fur	
silk	
paper	
cotton	
wood	
amber	
rubber balloon	(stronger tendency to gain electrons)
vinyl	
polyester	
ebonite	–

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